THE SEPARATION AXIOMS FOR MORPHISMS OF ALGEBRAIC SPACES ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement quantifies over all separation axioms listed in the definition; the proof works with a general property of the diagonal, so it covers the whole list at once.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The list of separation axioms in terms of the diagonal. Definition definition-separated, used at proof lines 490-491 and 500-502. Fixes the properties of the diagonal covered by the lemma.
- Identification of the diagonal after base change. used at proof lines 496-498. Writes Delta_{X'/Y'} as a base change of Delta_{X/Y}.
- Stability of representable properties of morphisms under base change. Section section-representable, used at proof lines 498-500. Gives stability of each property of the diagonal.

Source reference. Original definition 03HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4571–4571. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *All of the separation axioms listed in Definition 03HL are stable under base change.*

Proof. Let $f : X \to Y$ and $Y' \to Y$ be morphisms of algebraic spaces. Let $f' : X' \to Y'$ be the base change of f by $Y' \to Y$. Then $\Delta_{X'/Y'}$ is the base change of $\Delta_{X/Y}$ by the morphism $X' \times_{Y'} X' \to X \times_Y X$. By the results of Section 03HA each of the properties of the diagonal used in Definition 03HL is stable under base change. Hence the lemma is true. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.